

Harder negative sets lower apparent AUROC and raise measured model contribution: a three-protocol calibration across 94 ENCODE eCLIP datasets

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Abstract

Sequence models for RNA-binding proteins are evaluated against negative windows, and how those windows are chosen is usually treated as an implementation detail. Across 94 paired ENCODE eCLIP datasets we hold the model, the positives, the chromosome-blocked folds and the estimator fixed and vary only the negative-set protocol, measuring the nested contribution of a 4-mer model over a 19-feature mono- and dinucleotide composition baseline refit under each protocol. Replacing GC-matched with dinucleotide-matched negatives lowers apparent AUROC by 0.110 in 94 of 94 datasets while raising the measured contribution from 0.027 to 0.066. A bias-aware protocol drawing negatives from other proteins’ binding sites is the easiest of the three discriminations and yields the least, 0.012. Apparent difficulty and measured contribution therefore move in opposite directions, spanning 5.4-fold, and that span is about three times the gap between a k-mer model and a convolutional network on the same datasets. The composition baseline each protocol leaves behind accounts for most of the variation. No rescaling supplies a transportable remedy: an exponent that equalises our three protocols exists, but the exponent fitted to an independent benchmark differs from it 2.4-fold. We recommend reporting the composition-only AUROC obtained under the same protocol beside every headline AUROC, and not comparing contributions measured under different protocols.

Keywords: RNA-binding proteins; eCLIP; benchmarking; negative-set construction; sequence composition; incremental value; AUROC

1 Introduction

Predicting where an RNA-binding protein (RBP) binds from sequence alone is a standard supervised problem, and progress is measured almost entirely by AUROC on held-out windows. Positives come from a crosslinking assay, usually eCLIP [2, 3], and the negatives are constructed. Horlacher and colleagues [4] made the consequences of that construction explicit: across eleven RBP prediction methods and hundreds of CLIP-seq experiments, the apparent performance of a method depends on how its negative set was built, and they proposed a bias-aware alternative in which negatives are drawn from other RBPs’ binding sites rather than from genomic background. That the number moves is therefore established. What has not been established is what the number is a function of, how far it moves once a composition baseline is accounted for, and whether any reported quantity survives the change.

The question matters because a raw AUROC conflates two things a reader cannot separate: how much signal the model found, and how easy the discrimination was made. A window set whose

negatives are sampled uniformly from the genome differs from its positives in base composition alone, and a model that detects “G-rich, C-poor” will score well without having learned anything sequence-specific. The standard mitigation is to match negatives on composition, most often GC content. But GC content is one linear functional of the sixteen-cell dinucleotide simplex, so matching on it leaves fifteen degrees of freedom free, and the residual compositional difference is available to any model. This is the same structure that Grimm and colleagues [6] identified in variant-effect benchmarks, where the way negatives and positives are assembled produces circularity that inflates apparent method performance and reorders methods, and it is a recognised failure mode of genomic machine-learning evaluation more generally [6, 7].

The natural response is to report not the model’s AUROC but what it adds over an explicit composition baseline. That quantity, which we call the nested contribution, has an exact operational meaning: fit the composition features alone and then with the model’s score, on the same rows and the same folds, and take the difference in out-of-fold AUROC. It is the quantity a reader wants when asking what a model contributes, and it is the quantity we measure here.

Our contribution is a calibration. We take 94 ENCODE eCLIP datasets, hold the model, the positives, the chromosome-blocked folds and the estimator fixed, and build negatives three ways: matched on GC content, matched on all sixteen dinucleotide frequencies, and drawn from other RBPs’ binding sites in the same cell line following the bias-aware recipe of Horlacher et al. [4]. Under each protocol we refit the composition baseline on that protocol’s own windows, so the baseline is never imported across designs. Four results follow.

First, apparent difficulty and measured contribution move in *opposite* directions. Dinucleotide matching lowers apparent AUROC in every one of the 94 datasets while roughly tripling the measured contribution, and the bias-aware protocol, which the field reads as the most principled of the three, turns out to be the *easiest* discrimination and yields the *least* measured contribution. A reader who infers from a lower score that a benchmark is more demanding, or from a higher score that a model is better, can be wrong in a direction that no amount of care with the model can fix.

Second, the range is large in the units that matter. The same 4-mer on the same positives measures 0.066, 0.027 or 0.012 AUROC of contribution depending only on the negatives, a 5.4-fold span, and the absolute spread of 0.054 is about three times the 0.019 that separates a 4-mer logistic regression from a convolutional network on our own panel. We are deliberate about which comparison this licenses. Protocol choice exceeds the difference between adjacent conventional model classes, so a benchmark that changes its negatives can reorder such methods. It does not exceed every architectural difference: the step from the convolutional network to a fine-tuned nucleotide language model is 0.103 on the same panel, about twice the protocol spread. The claim is that protocol effects are of the same order as, and often larger than, the effects these benchmarks are built to resolve, not that they dominate all of them.

Third, the composition baseline each protocol leaves behind accounts for most of the variation, and we report that mechanism rather than defending against it. Given the baseline, knowing which protocol produced it adds about one per cent of variance; given the protocol, knowing the baseline adds eleven. The relationship is not uniform, however: it is a property of composition-matched negatives and is absent for the bias-aware arm, so there are two protocol families rather than three interchangeable protocols, and a residual difference between families survives at matched baseline. That family structure replicates on an independent benchmark.

Fourth, no rescaling supplies a transportable fix. AUROC is compressive near one, so part of the span is arithmetic, and eight standard monotone reparameterisations reduce it. None removes it: the smallest span over the eight is 2.00-fold, far outside the range expected if the three protocols had equal true means. An exponent that does equalise our three protocols exists, but it is fitted, and the exponent fitted to an independent benchmark differs from ours by a factor of 2.4, leaving that

benchmark essentially where it started. The absence of a protocol-free measure is thus a statement about transportability, which is testable, rather than about a floor, which is not.

We therefore recommend a two-number report: state the composition-only AUROC obtained under the same protocol beside every headline AUROC. We are explicit about what this does and does not buy. It makes the problem visible and it is the coordinate least sensitive to protocol among those we examined, but on an independent benchmark it does not improve cross-protocol agreement, so we do not claim it makes contributions comparable. It does not; we recommend not comparing them.

Throughout, we use “measured contribution” and never “true contribution”. The three protocols do not estimate a common quantity, so words such as hidden, concealed, inflated or proper would presuppose a value this work argues does not travel between benchmarks. The register is calibration.

2 Results

2.1 Apparent difficulty and measured contribution move in opposite directions

Holding the model, the positives, the folds and the estimator fixed and changing only how the negative windows were built gives three different pictures of the same 4-mer on the same 94 datasets (Table 1, Figure 1).

Table 1: The same model on the same positives under three negative-set protocols, $n = 94$ datasets. Only the construction of the negative windows differs.

protocol	composition alone	apparent AUROC	nested contribution
dinucleotide-matched	0.6274	0.6937	+0.0663
GC-matched	0.7827	0.8092	+0.0265
bias-aware (other RBPs’ sites)	0.8248	0.8370	+0.0122

Moving from GC-matched to dinucleotide-matched negatives lowers apparent AUROC by 0.1095, and it does so in 94 of 94 datasets (paired Wilcoxon $p < 10^{-15}$). The same change raises the measured contribution from +0.0265 to +0.0663, a contrast of +0.0398 [+0.0325, +0.0477] (protein-clustered, positive in 88 of 94). A reader comparing the two benchmarks by headline AUROC would conclude the model had got worse. A reader comparing what the model added would conclude it had got nearly three times better. Both readings are correct about their own quantity, and they point in opposite directions.

The bias-aware protocol is where this becomes a problem for practice. Drawing negatives from other proteins’ binding sites is designed to remove the compositional and accessibility artefacts that genomic-background negatives carry, and it does: those negatives are transcribed, crosslink-accessible and strand-correct by construction, and no composition matcher touches them. We pre-registered the prediction that composition would fall toward 0.5 under this protocol, because both classes are real crosslink sites, and that the measured contribution would rise. Both halves were wrong. Composition alone is the *highest* of the three arms at 0.8248, the discrimination is the *easiest* of the three by apparent AUROC at 0.8370, and the measured contribution is the *lowest* at +0.0122, 0.53 times the GC arm (geometric mean over the 83 datasets positive in both arms; the ratio of panel means is 0.46). The reason is visible in hindsight and is itself a result: different RBPs bind compositionally different sites, so telling one protein’s sites from another’s is largely a composition task.

The ordering is therefore monotone in measured difficulty and not in how principled the protocol sounds. Ranked by composition alone, by apparent AUROC or by contribution, all three orderings agree. “Harder negatives reveal more of what a model adds” is exactly true. What is false is the inference that a bias-aware protocol is harder: it is the easiest of the three, and it reveals the least.

Two further features of Table 1 matter for interpretation. Composition alone beats the 4-mer outright in 29 of 94 datasets under GC matching. And the 4-mer’s contribution is significantly positive in 80 of 94 datasets, falling to 72 of 94 at the design effect measured for these data (Methods), so on roughly a quarter of the panel the model adds nothing detectable over composition under the protocol in common use.

The *sign* of the GC-versus-dinucleotide contrast is implied by the design, and we state this here rather than in a limitations paragraph. The composition baseline occupies the fifteen degrees of freedom of the dinucleotide simplex; GC matching constrains one linear functional of that space and dinucleotide matching constrains all fifteen, so the arm that controls more of the baseline must leave the baseline weaker and the model more room. Only the *magnitude* is informative. What the design does not imply is the bias-aware arm’s position: that protocol constrains *none* of the fifteen degrees of freedom, its negatives are not composition-matched in any respect (median absolute GC gap 0.1387 against the GC arm’s 0.0297), and it nonetheless has the highest composition baseline of the three. No degrees-of-freedom argument predicts that, and it is what breaks a purely circular reading of the table.

Artefact controls. Two alternative explanations were pre-registered and bounded against matched placebos, each with twenty placebo seeds. A strand-annotation artefact, arising because a matched negative inherits its positive’s strand label, accounts for -0.0055 $[-0.0089, -0.0022]$ of the contrast, leaving 85.4% standing. Negatives drawn from untranscribed loci account for -0.0043 $[-0.0069, -0.0018]$, leaving 89.7%; 40.1% of negatives lie in untranscribed loci, but that fraction is balanced across the two arms to within 1.2×10^{-5} ($p = 0.64$), and a confound present equally in both arms cannot manufacture a difference between them. One argument needs no placebo: GC-matched negatives are 42.8% sense against 47.4% in the dinucleotide arm, higher in 37 of 40 datasets ($p = 5.0 \times 10^{-9}$), so the spurious directional cue is stronger in the arm with the *smaller* contribution and works against the reported direction.

2.2 No rescaling supplies a transportable protocol-free quantity

AUROC is compressive near one, so a gain measured against a 0.82 baseline is arithmetically smaller than the same gain against a 0.63 baseline, and part of the 5.42-fold span is scale rather than protocol. We searched for a coordinate in which the span disappears (Figure 2).

Over eight monotone reparameterisations of the same 282 protocol-dataset cells, the span falls but does not close. Somers’ D [10] is an affine map of AUROC and returns 5.42 by construction, which serves as an internal control on the sweep. Every unbounded link reduces the span, in the order of how aggressively it stretches the upper tail: arcsine 3.99, complementary log-log 3.94, binormal d' 3.18, logit 2.72. The smallest is obtained by dividing the contribution by the baseline’s own headroom, 1 minus the composition AUROC, at 2.00 [1.67, 2.46]. Normalising instead by the excess over chance, which is the transformation a reader is most likely to invent, makes the span far worse at 18.2, because the denominator itself varies with protocol and approaches zero.

The 2.00 figure should not be compared against 1.0. A ratio of the largest to smallest of three noisy means is bounded below by one and biased upward, so some span is expected under any sampling. Simulating equal true arm means while preserving the real between-arm dataset pairing

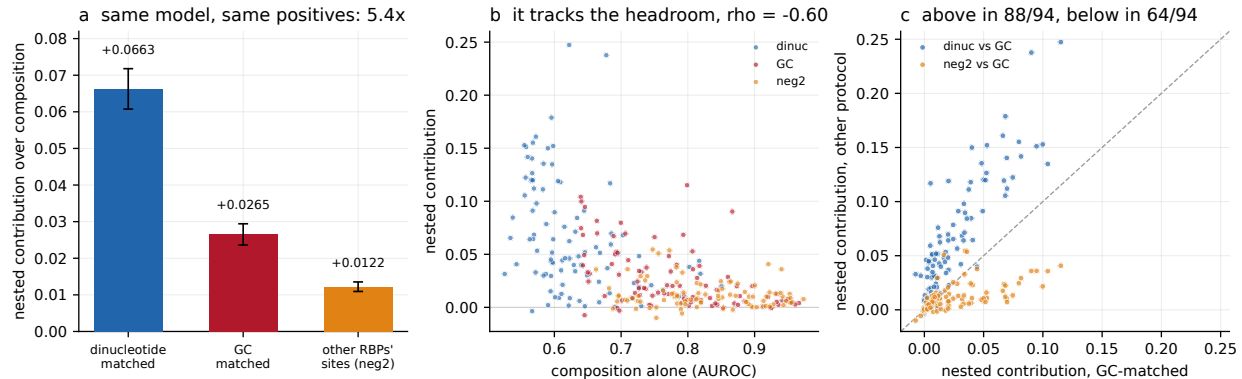


Figure 1: **The three protocols, and the reversal.** Same 94 datasets, same positives, same 4-mer, same folds and same estimator; only the negatives differ. (a) Composition-only AUROC, apparent AUROC and nested contribution under each protocol. (b) Nested contribution per dataset, dinucleotide-matched against GC-matched, paired; points above the diagonal are the 88 of 94 datasets where the model contributes more under dinucleotide matching. (c) The same for the bias-aware arm against GC-matched, lower in 64 of 94. Apparent AUROC moves in the opposite direction to contribution: dinucleotide matching lowers it in 94 of 94 datasets.

gives a null with median 1.064 and 95th percentile 1.175. The observed 2.00 lies well outside it, so the residual span is not an artefact of taking a ratio.

The headroom coordinate is not a coordinate, however, but a fitted member of a family, and we report that rather than resting on it. Dividing by headroom is the $p = 1$ case of dividing by headroom raised to the power p , and our sweep stopped at exactly the member we recommend. Extending the family, the span collapses:

Table 2: Fold span across the three protocols for $g/(1 - c)^p$, where g is the nested contribution and c the composition AUROC. An equalising exponent exists in sample.

p	0	0.5	1.0	1.25	1.544	1.75	2.0
span	5.42	3.43	2.00	1.48	1.005	1.34	1.95

At $p = 1.544$ the three protocols agree on the panel mean to within 0.5%. A monotone rescaling that equalises our three protocols therefore exists, and any claim that none does would be false on our own data.

The exponent is a property of the benchmark rather than of the quantity, and that is the result. Fitted to the two negative sets released by Horlacher et al. [4], the equalising exponent is 3.649, a factor of 2.4 from ours. Our exponent leaves their benchmark at 2.340 against the 2.381 it started from, that is, it buys nothing there. So a normalisation strong enough to equalise one benchmark must be refitted on the next, and the absence of a protocol-free measure is a statement about transportability rather than about a floor. Stated that way it is testable, and we have tested it on data we did not build.

2.3 It is the composition baseline, not the protocol label

Pooled across all 282 protocol-dataset cells the contribution tracks the composition baseline at Spearman -0.600 ($p = 6 \times 10^{-29}$). We report this as the mechanism rather than defending against

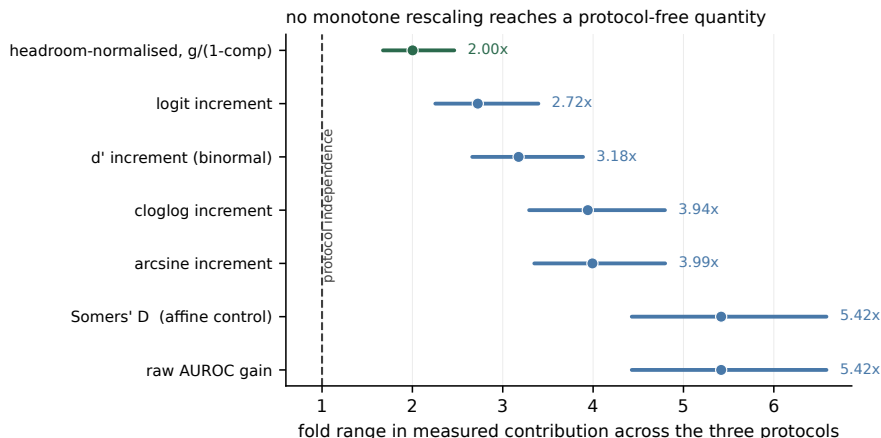


Figure 2: **No rescaling reaches a protocol-free quantity among fixed coordinates.** Fold span in measured contribution across the three protocols under eight monotone reparameterisations of the same 282 protocol-dataset cells. Points are the span, bars are 95% percentile intervals from a bootstrap over datasets (2000 draws), and the dashed line marks protocol independence. Somers' D is an affine map of AUROC and returns the raw value by construction, serving as a control on the sweep. The minimum, 2.00 [1.67, 2.46], is attained by dividing by the baseline's own headroom.

it: most of what a protocol does is decide how much room the baseline leaves. Three analyses localise the effect (Figure 3).

Variance accounting. Regressing the contribution on a cubic in the composition baseline over all 282 cells and then adding protocol dummies, the protocol label adds 1.0% of variance given the baseline, while the baseline adds 11.0% given the label: an order of magnitude in the baseline's favour.

A natural experiment. The bias-aware protocol usually raises the baseline relative to GC matching, but in 27 of 94 datasets it lowers it. If the protocol label carried the information, its deficit would persist there. It reverses: -0.0212 [-0.0269 , -0.0157] where it raises the baseline, $+0.0028$ [-0.0000 , $+0.0063$] where it lowers it, with within-dataset Spearman correlation between the change in baseline and the change in contribution of -0.664 ($p = 3 \times 10^{-13}$). Whichever protocol leaves the lower baseline yields the higher contribution, regardless of which protocol it is.

Matching on the baseline. Because the dinucleotide baseline is lower than the GC baseline in 94 of 94 datasets, arm and baseline are perfectly rank-confounded within a dataset and the comparison must borrow across datasets. Doing so, the headline $+0.0398$ falls to -0.0087 [-0.0265 , $+0.0122$]. We report this and also report why it is not a clean test: the 41 retained pairs are selected from opposing tails, the dinucleotide members having mean baseline 0.6875 against the panel's 0.6274 and the GC members 0.6904 against 0.7827; only 24 distinct GC partners serve 41 comparisons, with one reused eleven times; and no protein matching is possible. Nearest-neighbour matching on a variable monotonically related to the outcome, with reuse and across proteins, is biased toward zero by construction here. The honest reading is that the comparison is not identified for this pair, which is consistent with the pair's common support being only 0.073 AUROC wide over 33 of 188 cells.

Where it is identified, a protocol-family residual survives. The GC-versus-bias-aware pair overlaps by 0.212 AUROC over 130 of 188 cells, so the same matched design is straightforward there. Under two independent designs the bias-aware arm still costs -0.0081 $[-0.0130, -0.0036]$ (nearest-baseline matching) and -0.0043 $[-0.0077, -0.0012]$ (within-dataset intercept at zero baseline shift). Decomposed by pair, the protocol label’s incremental variance is 0.05% for GC versus dinucleotide but 3.40% for GC versus bias-aware, so the pooled 1.0% averages a pair where the question is unanswerable with one where it is answerable.

The mechanism, which changes the framing from three protocols to two families. The gradient from baseline to contribution is a property of composition-matched negatives and is absent otherwise: within-arm Spearman is -0.545 (GC), -0.462 (dinucleotide) and -0.122 , not significant (bias-aware). Inside the composition-matched family the baseline is essentially the whole story; across families it is not, and a residual of about 0.005 to 0.008 AUROC remains. This is why we do not name headroom as the general mechanism.

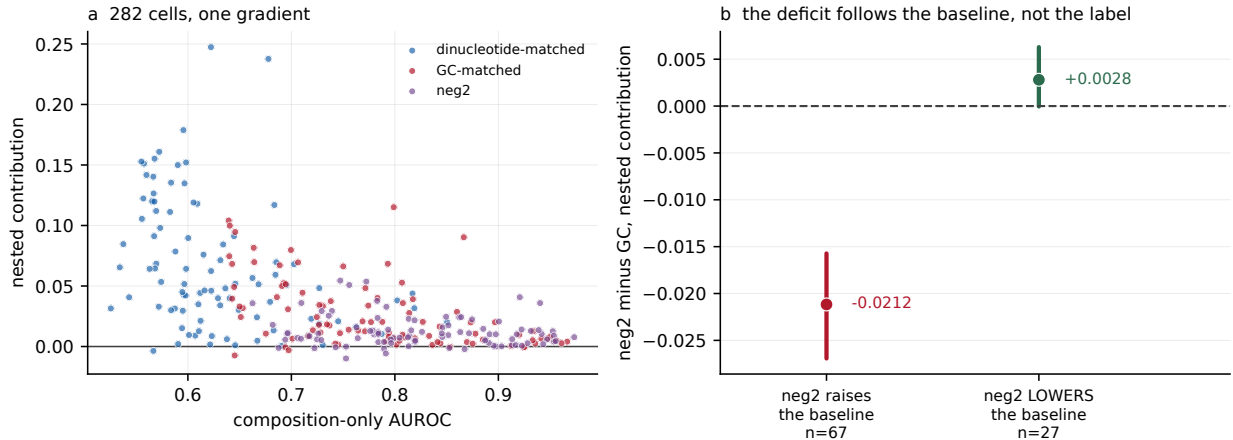


Figure 3: **The composition baseline, not the protocol label.** (a) All 282 protocol-dataset cells: nested contribution against the composition-only AUROC the protocol left behind, coloured by protocol; pooled Spearman -0.600 ($p = 6 \times 10^{-29}$). (b) The natural experiment. Datasets are split by whether the bias-aware protocol raised or lowered the composition baseline relative to GC matching ($n = 67$ and 27); points are stratum means and bars are 95% percentile intervals from a bootstrap over datasets (4000 draws). The deficit reverses where the baseline falls.

2.4 The effect is not a property of the model class

The GC arm was swept for two further architectures, 470 fold-runs each: a 7089-parameter convolutional network in the style of Alipanahi et al. [13] and a fully fine-tuned 19.78 M-parameter SpliceBERT [15], using the same folds, the same seed and the same code path, with per-window scores committed (Figure 4).

The contrast is present for all three model classes on all 94 datasets, which retires the objection that a five-fold span measured with a bag of k -mers is an artefact of a weak model: it is still a factor of 2.4 for a fine-tuned transformer measured on the same rows. On the ratio scale the ordering reverses, the multiplier being 3.08, 3.51 and 2.38 for the three models (geometric mean over the 77 datasets positive in both arms for all models), so the contrast does not grow with capacity and we make no such claim.

Table 3: The protocol contrast for three model classes, $n = 94$ datasets, identical rows and folds. Intervals are protein-clustered percentile intervals over the 79 proteins.

model	GC arm	dinucleotide arm	contrast
4-mer	+0.0265	+0.0663	+0.0398 [+0.0325, +0.0477]
CNN	+0.0330	+0.0860	+0.0530 [+0.0440, +0.0627]
SpliceBERT	+0.0890	+0.1754	+0.0864 [+0.0778, +0.0951]

Whose property is the multiplier? Decomposing its log over the 262 dataset-by-model cells where both arms are positive, RBP identity accounts for 64.8% of the variance, cell line 0.2% and model class 2.8%. Those shares are not directly comparable, because a factor with 79 levels absorbs 29.7% of the variance of relabelled data while a three-level factor absorbs 0.8%. Measured against a null that permutes protein labels between datasets while keeping each dataset-by-model block intact, protein’s excess is +7.2 points ($p = 0.003$), and the direct test is stronger than either share: the same protein’s log multiplier agrees across the two cell lines at $r = 0.586$ (Spearman 0.594, $p = 0.0001$) over 40 protein-by-model pairs. Cell line is not detectable, though with only fifteen proteins in both lines the design has little power to detect it. Model class is small but real ($p = 0.023$), consistent with SpliceBERT’s significantly lower multiplier, and it survives the differential censoring visible in its own input.

Why we do not decompose the span into compression and a residual protocol effect.

The natural next step is to transplant a model’s discriminability increment across baselines and call the unexplained part a protocol effect. That quantity is not identified for any of the three models. Its central assumption, that the increment is invariant to the baseline it is measured against, is violated in these data, and across a grid of twelve reasonable specifications only 4 of 6 sign tests hold for the 4-mer, 6 of 6 for the CNN and 1 of 6 for SpliceBERT. One standard link, the log odds, reverses the sign entirely, which is the behaviour Mood [12] describes for odds ratios under unobserved heterogeneity. With three arms the transplant residual’s sign follows the direction of transport: carrying an increment from a high-baseline arm to a low-baseline one gives +0.0452 and the reverse gives -0.0259 . We therefore report the raw contrast, which needs no transplant, no link and no transportability assumption, and treat the decomposition as a sensitivity analysis whose failure is itself informative.

A provenance limitation on this section specifically. For 20 of the 94 datasets in the dinucleotide arm, the committed neural-network scores were produced under a partition that is not chromosome-grouped: fold sizes were preserved, so no count-based check detects it, but up to 23 chromosomes appear in a single fold and up to 44.5% of rows have a same-strand neighbour within 1 kb assigned to a different fold, against exactly zero under the study folds and zero throughout the GC arm. The 4-mer is refit on the study folds for every dataset and so is unaffected, which makes it a clean internal control. Restricting the neural comparison to the 74 correctly partitioned datasets gives contrasts of +0.0436 (4-mer), +0.0506 (CNN) and +0.0878 (SpliceBERT), all within the protein-clustered interval half-widths of Table 3. We report the 74-dataset figures as primary for the two neural models and the 94-dataset figures as a sensitivity.

A second asymmetry in the same section, disclosed for completeness: the 4-mer entered the nested fit as a log odds and the neural models as probabilities, and a logistic regression is not invariant to a nonlinear transform of a covariate. Refitting the neural arms on the logit scale raises their contributions by +0.0008 (CNN) and +0.0030 (SpliceBERT) on the GC arm, so the

published values are conservative, and moves the contrasts by at most 0.0004. AUROC is invariant to monotone transforms, so no standalone AUROC is affected.

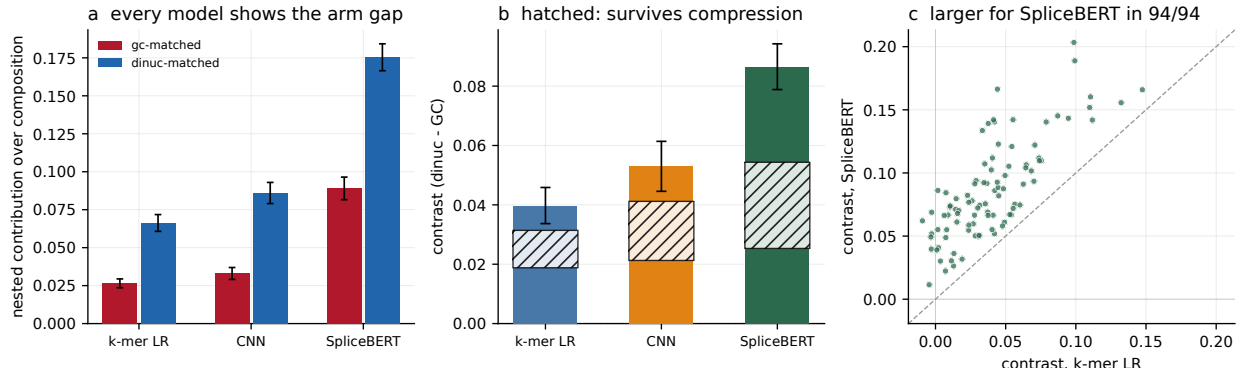


Figure 4: **The contrast holds for three model classes.** Nested contribution in each arm for a 4-mer logistic regression, a 7089-parameter convolutional network and a fully fine-tuned 19.78 M-parameter SpliceBERT, on identical rows and folds, $n = 94$. Intervals are 95% percentile intervals from a bootstrap over the 79 proteins, taking all datasets of each sampled protein (4000 draws). On the ratio scale the ordering reverses, so the contrast does not grow with capacity.

2.5 The calibration replicates on an independent benchmark

Every result above is measured on windows we built. To test whether any of it travels we applied the identical decomposition to the negative sets released by Horlacher et al. [4]: their positives, their peak calling, their negative-set construction and their fold assignments, over the 45 datasets shared with our panel. Only the measurement is ours (Figure 5).

Table 4: The same decomposition on an independent benchmark, $n = 45$ datasets. Their positives, negatives and folds.

their negative set	composition	apparent	contribution
negative-1 (transcript background)	0.8211	0.8606	+0.0395
negative-2 (other RBPs' sites)	0.7560	0.7726	+0.0166

The span on their benchmark is 2.381, against 2.17 for the family-matched pair in our own data, the GC versus bias-aware comparison; their negative-2 is constructed like our bias-aware arm, so that is the correct comparison rather than our dinucleotide-versus-GC 2.50.

More importantly, the two-family mechanism replicates arm for arm. The within-arm gradient from baseline to contribution on their data is -0.645 ($p = 1.8 \times 10^{-6}$) for their composition-matched transcript-background arm and -0.187 ($p = 0.22$, not significant) for their other-RBPs'-sites arm, against -0.545 and -0.462 for our composition-matched arms and -0.122 , not significant, for ours. An independent laboratory's negatives and folds reproduce both the presence of the gradient in one family and its absence in the other.

This also explains the one feature of their benchmark that is anti-monotone to ours: their negative-2 has both the lower baseline and the lower contribution, which no single-family headroom relation predicts and which a cross-family residual does.

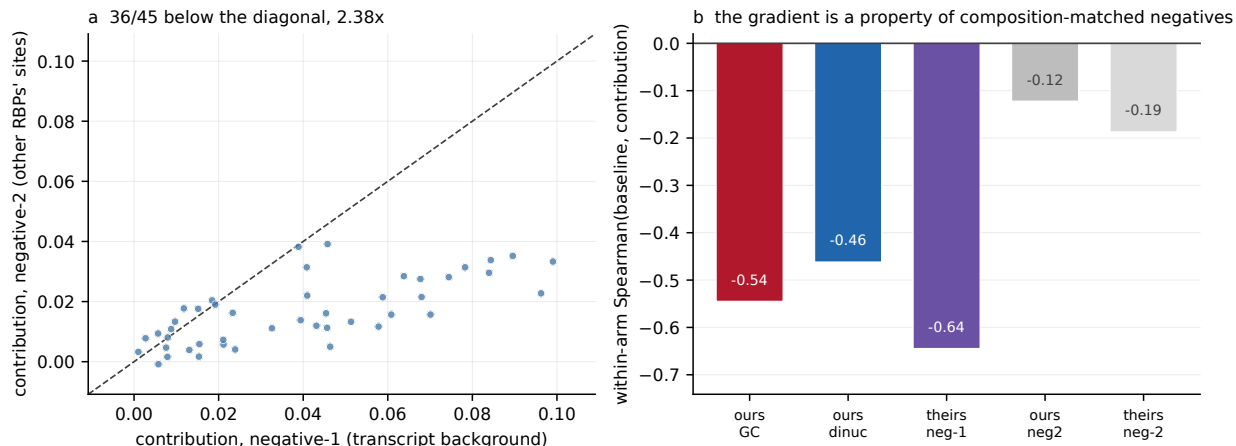


Figure 5: **Replication on an independent benchmark.** Horlacher et al. [4]’s released negative sets, their peak calling and their folds, over the 45 datasets shared with our panel; only the measurement is ours. (a) Nested contribution under their negative-2 against their negative-1, paired per dataset; 36 of 45 fall below the diagonal. (b) The mechanism, which is what travels: the within-arm gradient from composition baseline to contribution, for the composition-matched arms in both benchmarks and the other-RBPs’-sites arms in both. The gradient is present in one family and absent in the other, in both benchmarks independently.

2.6 The recommendation, and the limits of what it buys

The deliverable is a two-number report: state the composition-only AUROC obtained under the same protocol beside every headline AUROC. Everything above shows that not doing so is a problem. Whether doing so helps is a separate question, and we tested it by putting a reader in the position the recommendation is meant to rescue, holding two papers that used different protocols and wanting to compare what the models contributed (Figure 6).

On our data, normalising by headroom improves rank agreement in 3 of 3 protocol pairs and reduces disagreement on a common scale in 3 of 3, by 58%, 10% and 48%. Rank agreement is scale-free, so no normalisation can flatter it.

Two findings qualify this. First, only the GC-versus-dinucleotide rank improvement has an interval clear of zero (+0.051 [+0.007, +0.115]); the other two pairs are consistent in direction and individually null. That pair is precisely the one where the protocol label adds 0.05% of variance, so the fix is best demonstrated where it is least needed. Second, and decisively, the test fails to replicate out of sample. On the 45 datasets of Horlacher et al. [4], rank agreement *falls* from 0.706 to 0.656 under normalisation and disagreement *rises* from 0.860 to 0.908. Those are, verbatim, the two criteria we pre-registered as falsifying. The change in rank agreement is -0.050 [-0.222 , $+0.140$], so at $n = 45$ this is a failure to replicate rather than a refutation, but it points the wrong way and we report it.

We therefore make the weaker claim that the evidence supports. Report the composition-only AUROC under the same protocol, because it is the only summary of what the protocol did that a reader can act on and because it makes the problem visible. Do not treat any normalisation as making contributions comparable across protocols: we cannot demonstrate that it does, and on independent data it does not. The operational recommendation is to report the baseline and to refrain from cross-protocol comparison, not to rescale and proceed.



Figure 6: **The recommendation, and where it stops working.** (a) On our data, cross-protocol rank agreement before and after dividing by the baseline’s headroom, for each protocol pair. Agreement improves in 3 of 3, but only the GC-versus-dinucleotide improvement has a protein-clustered interval clear of zero. (b) The same test on the independent benchmark of Figure 5, $n = 45$: rank agreement falls and disagreement rises under normalisation, which are the two criteria pre-registered as falsifying the recommendation. The change in rank agreement is -0.050 $[-0.222, +0.140]$, so this is a failure to replicate rather than a refutation.

3 Discussion

The practical consequence of these measurements is narrow and actionable. When reporting how well a sequence model identifies RBP binding sites, report the composition-only AUROC obtained under the same protocol beside the headline AUROC, and do not compare contributions measured under different negative-set protocols. The first is cheap: it is one extra logistic regression on nineteen features. The second is a restriction on inference rather than an addition to it, and it is the part that costs something, because much of the field’s cross-paper comparison is of exactly that kind.

Three things follow that are worth separating from the recommendation itself.

Apparent difficulty is not a proxy for benchmark quality. The intuition that a lower score on a harder negative set indicates a more honest evaluation is what the reversal breaks. Moving to dinucleotide-matched negatives lowers apparent AUROC in every dataset we examined while raising the measured contribution nearly threefold, and the bias-aware protocol, which removes real artefacts and is well motivated on those grounds, produces the easiest discrimination and the smallest measured contribution of the three. Nothing about the bias-aware design is wrong; what is wrong is reading its scores as information about how much a model has learned. Different proteins bind compositionally different sites, so separating one protein’s sites from another’s is largely a composition task, and a protocol that makes that the task leaves a sequence model little to add.

There are two protocol families, not a continuum of stringency. The relation between the composition baseline a protocol leaves and the contribution a model then shows is strong for composition-matched negatives, in our data and independently in Horlacher et al. [4]’s, and absent for negatives drawn from other proteins’ sites. A residual difference between the families survives matching on the baseline. So the useful taxonomy for a benchmark builder is not “how strict is the

matching” but “what kind of object is the negative”, and results are comparable within a family in a way they are not across families.

A quantity’s protocol dependence can be reduced by fitting and not by choosing. Eight standard reparameterisations shrink the span without closing it, and the best of them, dividing by the baseline’s headroom, is the two-number report expressed as one number. But an exponent that closes the span entirely exists, and it does not transport: the exponent fitted to our three protocols leaves an independent benchmark almost exactly where it started, and the exponent fitted to that benchmark is 2.4 times ours. This is the sense in which there is no protocol-free measure of what a sequence model contributes. It is not that no transformation works; it is that the transformation that works has to be refitted per benchmark, which is precisely what a comparable measure cannot require.

On what “adds over composition” means. The composition baseline stops at order two, which is what a dinucleotide-preserving shuffle would hold fixed and is a defensible place to stop. It is nonetheless a choice, and it does substantial work. Refitting the baseline at order three, adding the 63 trinucleotide frequencies, removes about four fifths of the 4-mer’s measured contribution and 87% of the headline contrast, and on the bias-aware arm the median dataset retains nothing at all. For a bag-of- k -mers model, most of what we and others report as “what the model adds over composition” is one further order of composition, not motif recognition and nothing positional. Any paper reporting an increment over a composition baseline is reporting a quantity indexed by that baseline’s order, and almost none state which order they used.

Two things temper that, and one of them is a result in its own right. The protocol dependence is not an artefact of where the baseline stops: at an order-three baseline the contrast remains +0.0067, +0.0309 and +0.0545 for the three model classes, all clear of zero, and the span across protocols does not collapse. And the collapse is specific to models whose features are nested inside the baseline’s span. Over a trinucleotide baseline the 4-mer’s contribution is positive in only 65 of 94 datasets while SpliceBERT’s is positive in 94 of 94, and once the raised baseline’s reduced headroom is accounted for the 4-mer loses about 1.4 times what SpliceBERT does. A benchmark that wants to separate model classes rather than rank them by how much short-range composition they absorb could therefore raise the baseline’s order. We state the circularity that recommendation carries: trinucleotide counts are linear aggregates of 4-mer counts, so an order-three baseline is structurally unfavourable to k -mer models, and it separates models by how much of their signal lies outside short-range composition rather than by capacity.

Relation to prior work. Horlacher et al. [4] established that negative-set construction changes apparent performance across many RBP methods and proposed the bias-aware alternative. This work does not contest that and depends on it: their released negative sets are our only external validation. What we add is the calibration. The quantity that moves is not just apparent AUROC but the increment over an explicit baseline; it moves in the opposite direction to apparent difficulty; it moves by a factor of five, comparable to or larger than the between-method differences these benchmarks exist to resolve; the composition baseline rather than the protocol label is what it tracks; and no reparameterisation makes it comparable across benchmarks. The same structural problem is documented in variant-effect prediction [6], where benchmark assembly produces circularity that both inflates and reorders methods, and the general sensitivity of genomic machine-learning evaluations to how the negative class is defined has been noted before [7]. Our contribution is to quantify it for one well-defined estimand in one well-covered assay, and to show which part of it is

arithmetic and which is not.

Limitations

The sign of the primary contrast is design-implied. GC matching constrains one of the composition baseline’s fifteen degrees of freedom and dinucleotide matching constrains all fifteen, so the direction of that contrast follows from the design and only its magnitude is informative. The bias-aware arm is not subject to this argument, since it constrains none of the fifteen and still has the highest baseline.

Every magnitude is indexed to a mono- and dinucleotide baseline. See above. The protocol dependence is not so indexed; the magnitudes are.

Achieved matching is not exact matching. The GC matcher’s operative acceptance bound is 0.15 rather than its nominal 0.05, holding the nominal tolerance for 94.8% of pairs, and the dinucleotide matcher reduces composition distance 2.27-fold rather than to zero. Degrees-of-freedom statements are exact for the design and approximate for the data.

Three protocols, two cell lines, three model classes, one assay. The three-protocol results use the 4-mer only, because the neural models were never trained on the bias-aware arm; the multi-model results use two arms only. The two halves of the argument therefore do not meet in a single cell, and extending the neural sweep to the third protocol is the most valuable single addition to this work.

Provenance limitations on the neural arms. For 20 of 94 datasets in the dinucleotide arm the committed neural scores came from a partition that is not chromosome-grouped; we report the 74 correctly partitioned datasets as primary and the full panel as a sensitivity, with shifts inside the interval half-widths. Initialisation was unseeded across the 940 fold-runs, at a measured cost of about 0.001 on a panel mean. The convolutional rung was trained on a different accelerator from the transformer rung. And the neural scores entered the nested fit on the probability scale while the 4-mer’s entered as a log odds, which understates the neural contributions by up to 0.0033 and moves the contrasts by at most 0.0004.

Positive sets differ slightly between arms. Window construction is identical, but a positive is dropped when its matcher finds no acceptable negative, and the two matchers fail on different windows. The two arms’ positive sets have a Jaccard similarity with median 0.9972 and minimum 0.9237, and are exactly identical in 10 of 94 datasets. The negatives are therefore almost, and not exactly, the only difference between arms.

No significance filter on positives and no expression filter on negatives. Every called peak is a positive, so the positive set includes weak peaks that a study applying ENCODE’s standard thresholds [3] would drop. The expression question is bounded by the control reported above rather than removed. Region labels are 6.1% to 8.7% wrong on the negative side by our own audit, which affects the region-matching step asymmetrically.

Two limitations of the verification harness itself. It asserts values, not the provenance of the code that produced them, and the fold-partition defect above is exactly the class of error that distinction permits: it was found by inspection, not by any of the 614 assertions. And its manuscript audit checks only numbers written to three or more decimal places, so a percentage written to one decimal is invisible to it and was checked by hand.

Inference is asymptotic in places. Confidence intervals on a single dataset’s contribution are Wald intervals on the DeLong standard error rather than bootstrap intervals, and the Firth coefficient intervals reported in the repository resample rows rather than genes, which is narrower than gene clustering would give.

4 Materials and Methods

4.1 Panel selection

Candidate experiments were every released ENCODE eCLIP experiment in K562 and HepG2 (139 and 105 respectively). Where a protein had several peak files we took the one with the most biological replicates, breaking ties by ENCODE’s `preferred_default` flag, and required `output_type=peaks`, `assembly=GRCh38`, `status=released` and `file_format=bed`. Every candidate was preprocessed under both composition-matching protocols; datasets retaining at least 400 out-of-fold scored pairs were eligible.

The study panel was then defined once by the dinucleotide arm, which is the stricter control: eligible datasets were sorted by pair count with a stable sort and every second one retained, giving 95 datasets over 79 proteins. Sampling systematically by rank rather than taking the largest N matters because AUROC correlates with dataset size at $r = 0.53$ to 0.67 in these data, so a size-truncated panel would confound the panel with the measured quantity; the retained panel spans the 0th to 99th percentile of the candidate size distribution. Of the 95, 94 also cleared the floor in the GC arm and constitute the paired panel used for every three-protocol result, giving 282 protocol-dataset cells; the model ladder uses all 95. The panel was written once to a manifest and every downstream stage reads it, so membership is not decided anywhere else. Supplementary Table S1 lists every dataset with its ENCFE file and ENCSR experiment accession.

Fifteen proteins are assayed in both cell lines and contribute two datasets each. All headline intervals therefore resample proteins rather than datasets.

4.2 Windows

Positive windows are 101 nt centred on the midpoint of each peak interval. Peak width and all significance columns are deliberately unused: no p -value, q -value or fold-change filter is applied, so the positive set is every called peak. Minus-strand windows are reverse-complemented and transcribed to RNA, so every sequence is the strand the protein sees. Windows containing an ambiguous base, falling outside the assembly, duplicating an existing window by start coordinate, or carrying no region annotation are dropped and counted. Analysis is restricted to chr1 to chr22 and chrX; chrY and all scaffolds and alternates are excluded.

Region class is assigned from GENCODE v45 [5] as the first of (5’UTR, 3’UTR, CDS, non-coding exon, intron) whose interval set contains the window midpoint. Region intervals are built from all transcripts with no gene-type filter, merged, and strand is deliberately discarded at this step; introns are computed as the gaps between consecutive exons. GTF 1-based inclusive coordinates are converted to 0-based half-open.

4.3 The three negative-set protocols

Each protocol produces one negative per positive. All three use a single random stream seeded at 7 per protein.

GC-matched. Candidates are windows of the same region class on the same chromosome, at least 500 nt from any peak of the target protein, drawn with probability proportional to the length of each free interval. A candidate is accepted if its GC content is within 0.05 of the positive’s. The matcher relaxes: after 25 unsuccessful draws the tolerance doubles to 0.10, and if 40 draws fail the best candidate seen is accepted provided it is within three times the nominal tolerance, that is 0.15. The operative acceptance bound is therefore 0.15, not 0.05, and we report achieved quality below rather than the nominal figure. Only the target protein’s own peaks are excluded, so a GC-matched or dinucleotide-matched negative may be another protein’s binding site. The negative inherits the positive’s strand label; the consequences are quantified in the Results.

Dinucleotide-matched. Positives are grouped by region and chromosome, and for each group a candidate pool of $\max(1500, 8n)$ windows is sampled. For each positive the nearest unused candidate is assigned greedily under the L_1 distance over the sixteen dinucleotide *counts*, searched with a k -d tree over the forty nearest neighbours. Counts rather than frequencies are used because integer distances are exactly representable, and each candidate additionally carries a tiebreak column that increases strictly with genomic position and is bounded below 0.5, so it can never reorder candidates whose integer distances differ but resolves exact ties by position. Without both devices the choice among tied candidates depends on the floating-point behaviour of the k -d tree implementation and a different negative is selected for about 8% of pairs between CPU architectures. Assignment is greedy rather than optimal; achieved distances are reported so the quality of the approximation is visible. Used coordinates are tracked across all groups, because one interval can be annotated as both a non-coding exon and a 3’UTR and so appear in two region pools. No maximum distance is imposed.

Bias-aware (other RBPs’ sites). Following Horlacher et al. [4], negatives for a target are the positive windows of the other panel proteins assayed in the same cell line, sampled 1:1 without replacement, excluding any donor window within 500 nt of any of the target’s own positive windows. Two deviations from Horlacher et al. [4] should be noted: our donor pool is the study panel (40 to 48 donor proteins per dataset, median 45) rather than a full ENCODE survey, and exclusion is by distance from the target’s 101 nt windows rather than from its full peak intervals. Sampling is performed within fold, so every pair stays inside one cross-validation fold. These negatives are transcribed, accessible to crosslinking and strand-correct by construction, and no composition matcher touches them; they are not composition-matched in any respect, which is central to interpreting the results.

4.4 Achieved match quality

Because the paper argues from what the matchers control, we report what they achieved over all 456,734 pairs per arm rather than their specifications.

Two points follow. The GC matcher holds its nominal tolerance for 94.8% of pairs and its observed maximum, 0.1486, sits at the 0.15 fallback cap. And “dinucleotide-matched” means a median L_1 of 0.220 rather than zero: the matcher improves composition distance 2.27-fold relative to the GC arm but does not eliminate it, so statements about degrees of freedom controlled are exact

Table 5: Achieved match quality between each positive and its assigned negative, all pairs. L_1 is over the sixteen dinucleotide frequencies, on a 0 to 2 scale.

arm	med. $ \Delta\text{GC} $	p90	p99	max	within 0.05	med. L_1
GC-matched	0.0297	0.0495	0.1089	0.1486	94.8%	0.500
dinucleotide-matched	0.0198	0.0693	0.1485	0.4159	85.0%	0.220
bias-aware	0.1387	0.3169	0.4555	0.6733	20.4%	0.700

for the design and approximate for the realisation. The bias-aware arm is not composition-matched at all, with a median GC gap 4.7 times the GC arm’s.

4.5 Cross-validation

Evaluation is grouped 5-fold cross-validation over whole chromosomes. The chromosome-to-fold map is frozen once for all datasets in both cell lines, so the fifteen proteins measured in both lines are evaluated on the same chromosomes and a between-line difference cannot be a partition artefact. The map was chosen by random-restart hill climbing on peak counts only, never on labels or model output, minimising the summed squared deviation from equal mass per fold with every dataset weighted equally; it was required to beat round-robin assignment, to hold the median per-dataset maximum deviation at or below 0.05, and to place no more than half of any dataset’s data in one fold. Fold i is the test fold, fold $i + 1 \bmod 5$ is validation and the remaining three train, so each fold is test once, validation once and training three times, and no additional data is spent on model selection.

Every window is scored exactly once out of fold and all AUROCs are computed on the pooled out-of-fold score vector. Pooling rather than averaging per-fold AUROCs makes the estimate invariant to fold size, which matters because the least balanced dataset places 0.42 of its data in a single fold.

Leakage between folds was audited by exact 32-mer sharing rather than by clustered identity: a 101 nt window contains 70 distinct 32-mers and two unrelated sequences share one with negligible probability, so a shared 32-mer is evidence of common origin.

4.6 The composition baseline and the nested contribution

The composition baseline is a 19-column design matrix: three mononucleotide frequencies, fifteen dinucleotide frequencies and the Shannon entropy of the window’s base composition. One level is dropped from each frequency family because frequencies sum to one and retaining all of them makes the design singular; dropping a level does not change the space the block spans. Every column is standardised. GC content is not a feature: it is the spanned combination C+G, and because mononucleotide counts are marginals of dinucleotide counts the block occupies the fifteen degrees of freedom of the dinucleotide simplex. The GC-matched protocol therefore constrains one linear functional of that space and the dinucleotide-matched protocol constrains all fifteen. The baseline is refit on each protocol’s own windows and is never carried between protocols.

The nested contribution of a model is

$$\text{AUROC}(\text{composition} + \text{standardised model score}) - \text{AUROC}(\text{composition alone}),$$

both fitted by L_2 -penalised logistic regression ($C = 1$, lbfgs, at most 3000 iterations) once per fold on that fold’s training rows, both evaluated as pooled out-of-fold linear predictors on identical rows

and identical folds, and compared by DeLong’s paired estimator [8] in the $O(n \log n)$ midrank form of Sun and Xu [9]. The paired estimator is required rather than optional: the two score vectors are fitted on overlapping training data and evaluated on the same rows, so they are strongly correlated and treating them as independent understates the variance of the difference. Confidence intervals on a single dataset’s contribution are Wald intervals on the DeLong standard error.

Both arms are evaluated out of fold. This is not a formality: an in-sample composition AUROC compared against an out-of-fold model AUROC flatters composition enough to reverse the comparison on individual datasets, reading as composition beating the model when it does not.

4.7 Models

4-mer logistic regression. Raw 4-mer counts over the RNA sequence (98 per window, 256 columns) into L_2 -penalised logistic regression ($C = 1$). One model per fold, out-of-fold score taken as the linear predictor. The k -mer order is 4, unrelated to the 5 in 5-fold cross-validation, and the contrast is positive at every order from 3 to 6.

Convolutional network. A DeepBind-style architecture [13]: convolution (4 to 16 channels, width 12), ReLU, max-pool 4, convolution (16 to 32, width 8), ReLU, global max-pool over position, dense 32 to 64, ReLU, dropout 0.5, dense 64 to 1. 7089 parameters, all trained from scratch. Global max-pooling makes the model position-invariant, which is a requirement rather than a convenience here: exploratory analysis placed the discriminative signal about 15 nt off centre, varying by protein.

SpliceBERT. The pretrained nucleotide language model of Chen et al. [15], `multimolecule/splicebert`, fully fine-tuned; 19.78 M parameters, all trainable. The classification head is a mean pool over real nucleotide tokens, masking classification, separator and padding tokens, then dense to 128, ReLU, dropout 0.3, dense to 1. Weights are baked into the container image at build time, so “which weights” has the same answer as “which image”.

Both neural models: AdamW, weight decay 0.01, batch size 32, binary cross-entropy, single precision, at most 12 epochs with early stopping on validation AUROC at patience 4, best checkpoint restored before test scoring. Learning rate 10^{-3} for the head and 3×10^{-5} for a fine-tuned encoder. Nominal epochs overstate the training actually performed: the median best epoch for SpliceBERT is 3 and most runs stopped by epoch 6 or 7, whereas the convolutional network typically ran the full 12. Initial weights were drawn before the seed was set, so initialisation is unseeded across the 940 committed fold-runs; the measured cost is a per-dataset standard deviation of 0.006 to 0.010 in the nested contribution, inducing about 0.001 on a 94-dataset mean against a reported interval half-width of about 0.008.

4.8 Inference

Because fifteen of the 79 proteins contribute two datasets each and the within-protein correlation of the primary contrast is 0.92, every headline interval is a percentile interval from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. Dataset-level intervals are 1.05 to 1.23 times narrower and are not reported as headline values.

Per-dataset significance counts are additionally reported at a design effect of 1.35 applied to the DeLong standard error. That factor is the product of two quantities measured on these data: 1.10 for DeLong conditioning on two score vectors as fixed when both are in fact fitted, and 1.23 for treating spatially clustered windows as independent, estimated by a 10 kb block bootstrap. It moves the count of datasets where the 4-mer significantly helps from 80 of 94 to 72 of 94.

4.9 Reproducibility

All published values are reproduced offline by a single command against committed evidence. The harness runs 614 numeric assertions against a frozen expectations file, and the number of assertions that ran is itself asserted, so a check cannot silently skip. Two assertions are stronger than regression gates: 285 published AUROCs are rebuilt from committed per-example scores to a maximum absolute difference of 2.2×10^{-16} , and the headline contrast is rebuilt from raw sequence to 1.2×10^{-6} . Per-window out-of-fold scores for all three model classes are committed, so every cell of the model-class comparison is recomputable from the repository alone rather than being asserted against a hand-entered table.

Three defences against platform-dependent results are worth naming because they are otherwise invisible: dinucleotide distances are computed on integer counts, candidate ties are broken by genomic position, and the panel sort is stable. Each was added after a specific observed discrepancy between CPU architectures or between runs.

Analyses used NumPy [16], SciPy [17], scikit-learn [18], PyTorch [19] and Matplotlib [20].

Data availability

All primary data are public. Positive windows derive from ENCODE eCLIP narrowPeak files [1, 3]; every dataset used, with its ENCFF file accession and ENCSR experiment accession, is listed in Supplementary Table S1 (95 datasets, 79 proteins, 95 experiments). Gene annotation is GENCODE v45 [5]. Cell-line expression used in the transcription control comes from four ENCODE polyA RNA-seq quantification files, accessions listed in the Supplementary Methods. The external validation uses the negative sets and fold assignments released by Horlacher et al. [4] at [doi:10.5281/zenodo.10600977](https://doi.org/10.5281/zenodo.10600977), downloaded and md5-verified.

Derived data supporting every number in this paper, including per-window out-of-fold model scores for all three model classes and all per-dataset result tables, are released with the code under CC BY 4.0. Intermediate window tables containing genomic sequence are not redistributed; they are regenerated from the accessions above by the pipeline in one command.

Code availability

All analysis code is available at <https://github.com/Nirmalkumar31/rbp-protocol-calibration> under the MIT licence, with `results/` and `data/evidence/` under CC BY 4.0. Every published value is reproducible offline from the repository with `python scripts/verify.py --local results/tables`, which runs 614 numeric assertions against a frozen expectations file and asserts the number of assertions that ran, so a check cannot silently skip.

Use of AI tools

Generative AI (Anthropic Claude) was used in the preparation of this work for software development, including the analysis and pipeline code in the accompanying repository, and for drafting and editing manuscript text. The study design, all analytical and interpretive decisions, and every scientific claim made here are the author’s own, and the author is solely responsible for the content of this manuscript and for the correctness of its results. All reported values are produced by code committed to the accompanying repository and are reproduced by its verification harness. No

citation or reference was generated by a language model; the reference list was verified against publisher records.

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Competing interests

The author declares no competing interests.

References

- [1] ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature*, 489:57–74, 2012. [doi:10.1038/nature11247](https://doi.org/10.1038/nature11247)
- [2] E. L. Van Nostrand, G. A. Pratt, A. A. Shishkin, et al. Robust transcriptome-wide discovery of RNA-binding protein binding sites with enhanced CLIP (eCLIP). *Nature Methods*, 13:508–514, 2016. [doi:10.1038/nmeth.3810](https://doi.org/10.1038/nmeth.3810)
- [3] E. L. Van Nostrand, P. Freese, G. A. Pratt, et al. A large-scale binding and functional map of human RNA-binding proteins. *Nature*, 583:711–719, 2020. [doi:10.1038/s41586-020-2077-3](https://doi.org/10.1038/s41586-020-2077-3)
- [4] M. Horlacher et al. A systematic benchmark of machine learning methods for protein-RNA interaction prediction. *Briefings in Bioinformatics*, 24(5):bbad307, 2023. [doi:10.1093/bib/bbad307](https://doi.org/10.1093/bib/bbad307)
- [5] A. Frankish, S. Carbonell-Sala, M. Diekhans, et al. GENCODE: reference annotation for the human and mouse genomes in 2023. *Nucleic Acids Research*, 51(D1):D942–D949, 2023. [doi:10.1093/nar/gkac1071](https://doi.org/10.1093/nar/gkac1071)
- [6] D. G. Grimm, C. A. Azencott, F. Aicheler, et al. The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation*, 36(5):513–523, 2015. [doi:10.1002/humu.22768](https://doi.org/10.1002/humu.22768)
- [7] S. Whalen, J. Schreiber, W. S. Noble, and K. S. Pollard. Navigating the pitfalls of applying machine learning in genomics. *Nature Reviews Genetics*, 23:169–181, 2022. [doi:10.1038/s41576-021-00434-9](https://doi.org/10.1038/s41576-021-00434-9)
- [8] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44:837–845, 1988. [doi:10.2307/2531595](https://doi.org/10.2307/2531595)
- [9] X. Sun and W. Xu. Fast implementation of DeLong’s algorithm for comparing the areas under correlated receiver operating characteristic curves. *IEEE Signal Processing Letters*, 21(11):1389–1393, 2014. [doi:10.1109/LSP.2014.2337313](https://doi.org/10.1109/LSP.2014.2337313)
- [10] R. H. Somers. A new asymmetric measure of association for ordinal variables. *American Sociological Review*, 27(6):799–811, 1962. [doi:10.2307/2090408](https://doi.org/10.2307/2090408)

- [11] D. Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993. [doi:10.1093/biomet/80.1.27](https://doi.org/10.1093/biomet/80.1.27)
- [12] C. Mood. Logistic regression: why we cannot do what we think we can do, and what we can do about it. *European Sociological Review*, 26(1):67–82, 2010. [doi:10.1093/esr/jcp006](https://doi.org/10.1093/esr/jcp006)
- [13] B. Alipanahi, A. DeLong, M. T. Weirauch, and B. J. Frey. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nature Biotechnology*, 33:831–838, 2015. [doi:10.1038/nbt.3300](https://doi.org/10.1038/nbt.3300)
- [14] D. Maticzka, S. J. Lange, F. Costa, and R. Backofen. GraphProt: modeling binding preferences of RNA-binding proteins. *Genome Biology*, 15:R17, 2014. [doi:10.1186/gb-2014-15-1-r17](https://doi.org/10.1186/gb-2014-15-1-r17)
- [15] K. Chen, Y. Zhou, M. Ding, et al. Self-supervised learning on millions of primary RNA sequences from 72 vertebrates improves sequence-based RNA splicing prediction. *Briefings in Bioinformatics*, 25(3):bbae163, 2024. [doi:10.1093/bib/bbae163](https://doi.org/10.1093/bib/bbae163)
- [16] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. [doi:10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- [17] P. Virtanen, R. Gommers, T. E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. [doi:10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- [18] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [19] A. Paszke, S. Gross, F. Massa, et al. PyTorch: an imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035, 2019.
- [20] J. D. Hunter. Matplotlib: a 2D graphics environment. *Computing in Science and Engineering*, 9(3):90–95, 2007. [doi:10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)